A Remote Sensing Framework for Automated Monitoring of Roadside Water Quality

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***Abstract*—While air pollution is the most visible environmental impact of transportation systems, water pollution and water quality issues are also of great importance in the transportation and environment nexus. However, how the transportation sector impacts the water quality of adjacent receiving water bodies is not fully understood or well studied. In this paper, we present our work in designing and implementing a cost-effective, portable system to provide continuous monitoring of roadside water quality. We have deployed the system at multiple locations within the Salmon Creek watershed to collect data, demonstrating the feasibility of using the system to provide long-term monitoring. Combined with advanced data analytics technologies, this system provides potentials to reveal and understand the correlation between transportation activities and water pollution parameters, which can effectively guide the design of the next-generation of environmental friendly and sustainable transportation systems.**

***Index Terms*—Remote Sensing; Environment Monitoring; Internet-of-Things; Big Data; Edge Computing**

I. Introduction

It is well known that the transportation sector is a major source of ambient air pollution, due to its contribution to elevated fine particulate matter (PM2.5), ozone, and nitrogen dioxide concentrations [1]. Numerous studies have been carried out to analyze and quantify the air pollution caused by the transportation systems [2]–[5]. While air pollution is the most visible environmental impact of transportation systems, water pollution and water quality issues are also of great importance in the transportation and environment nexus. Specifically, transportation systems can affect water quality directly in many ways, including stormwater runoff, deicing chemicals, vehicle exhaust, oil spills, and other pollutants. Since clean water is one of the most critical and scarce resources for human society, investigating the impact of the transportation sector on the water quality is a key to designing and developing the next-generation environmental responsible transportation systems. However, the impact of the transportation systems on water quality is not well studied or fully understood. Current practice of water quality assessment by transportation agencies entails periodical water sampling, which tends to miss short-lived events and presents challenges for remote and diffcult-to-access locations. Addressing these challenges require continuous and reliable data collection, as well as advanced data analytics techniques.

In this paper, we present our work in designing and developing a cost-effective, remote sensing framework to automate water quality monitoring (WQM), and incorporate data analytics and machine learning technologies to facilitate the informative data-driven decision-making process. Our system was tested in a lab setting with pollutants that are often seen near transportation systems. In addition, we have deployed the system in multiple locations in the Salmon Creek watershed to collect water quality data in the area, demonstrating the feasibility of using the system to enable long-term automated monitoring of roadside water quality. With the potential to make monitoring and analyzing the aquatic environment ubiquitous, this research opens up opportunities for facilitating decision-making with respect to addressing the long-existing water security issues caused by transportation systems, harnessing modern information and communication technologies that are capable of capturing and analyzing data at an unprecedented scale.

The rest of the paper is organized as follows. Section II reviews existing work in utilizing remote sensing and Internet-of-Things (IoT) technologies for WQM, as well as data analytics approaches for anlyzing the water quality data. Section III presents the detailed design of our remote sensing system. The system evaluation and deployment, as well as the data collection process are described in Section IV. Section V presents the potential data processing and analysis approaches, as well as the results. Finally, Section VI concludes the paper and proposes future directions.

II. Related Work

With the growing popularity and ubiquity of Industry 4.0 [6] and the digital transformation it enables, a variety of IoT applications have emerged to support various *smart* environments, fundamentally reshaping the way individuals live and work [7]. As an effort to address the environmental challenges human society is facing, applications of these modern technologies on sustainability has gained more and more attention [8]. In

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the last decade, remote sensing and data analytics technologies have been used to monitor water quality issues [9]–[11].

WQM faces several significant challenges: data communication, durability, reliability, power management, and cost. Relevant studies that address these challenges have been carried out in the past decade. In 2018, Beddows et al developed an Arduino-based underwater data logger [12] that addressed the latter 4 of the aforementioned challenges. The data logger survived underwater deployment in the Yucatan Peninsula for approximately two years. The authors achieved this with inexpensive components ($35 total), an energy-efficient workflow, and a robust device that can survive extended deployment in harsh conditions. The downside of the system is that it only monitors water flow, which restricts its sensor capabilities. Additionally, there is no wireless communication support. Expanding their data logger with new sensors and wireless communication would introduce new problems (e.g., calibration and interference to wildlife through wireless communication) and make the previous challenges more difficult.

Another Arduino based prototype was developed by Fonseca-Campos and et al. for monitoring seven different WQM parameters (dissolved oxygen, electrical conductivity, oxidation-reduction potential, pH, temperature, total dissolved solids, and turbidity) using commercial sensors [13]. Their experimental design was to monitor the water quality of a fish tank for 10 days and then compare the results to reference solutions and meters. The results of their research are reliable node deployment and successful use of wireless data communication (Zigbee). Power and long-term durability were not considered in their paper, and cost was not mentioned (an estimate for the total cost is between $300-$500).

In [14], the authors used frequency analysis to compare their Arduino-based WQM system, priced at $1040, to a $3400 commercial WQM system. The results were that the Arduino-based system gave acceptable results but would require further calibration in the post processing. Most of the challenges were not mentioned in this paper, but the calibration insight provided by the authors can assist in the further development of remote WQM.

In [15], a Wireless Sensor Network (WSN) based WQM system is used to collect water quality data in the Sitnika river in Kosovo, between May 2015 and January 2016. The water quality dataset contains about 26,000 samples; each sample contains a value for dissolved oxygen, pH, specific conductivity, and temperature. Additionally, the authors provide basic methods for wireless WQM data pre-processing, which has potential to give insight into the problems in remote data collection since this dataset reflects a real world WSN.

Existing work on WQM using sensing platforms mainly focus on water quality in general, and those systems were not designed specifically for analyzing the impact of transportation systems. In this paper, we present a remote sensing system which is designed, tested and deployed for analyzing transportation systems.

III. System Design and Integration

The overarching goal of this research is to design a remote sensing system which can be used to provide continuous monitoring of roadside water quality. To achieve this goal, several system characteristics need to be considered when making design decisions. First, the system should provide **multisensing** capabilities, so that various types of parameters can be measured simultaneously. Second, the system should be **energy-efficient** and have a sustainable power supply. Third, the system should have **communication** capabilities to transmit the data for further processing. Finally, the system should be **cost-effective** to enable scalability of the deployment.

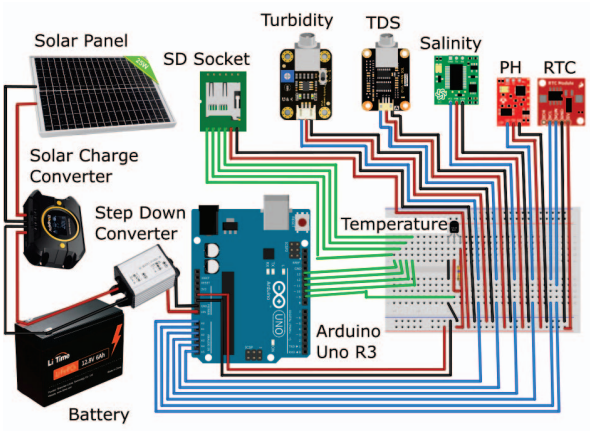


Fig. 1: Water Quality Monitoring System Architecture

To meet these requirements, we have designed and developed a cost effective, portable remote sensing platform to enable continuous monitoring on roadside water quality. The system architecture is shown in Figure 1. The system consists of sensing, power, and communication modules which are integrated together into a portable system which is easy to deploy. In this section, we describe in detail the individual system modules, as well as our approach to preparing the system for outdoor deployment.

A. System Modules

An Arduino serves as the central control component of the system. The Uno Rev3 model was chosen due to its low cost (about $28), small size, presence of enough pins to connect all needed components, sufficient memory and onboard storage to store and run our data collection script. This micro-controller was deemed capable enough for our purposes of periodically activating and recording sensor readings, making a more powerful microprocessor such as the Raspberry Pi unnecessary.

A select range of sensors were chosen to record both the physical and chemical properties of the water. Table I gives a short description of each sensor and their cost at the time of purchase. In total, all the sensors cost around $238. Many of these these same sensors were also used in Rao's and et Al. paper [14].

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TABLE I: The Physical and Chemical Sensors used in System

Type	Sensor	Measures	Vendor	Calibration	Price
Physical	Turbidity	How clear the water is	DFRobot	No calibration needed	$10
	Temperature	How hot or cold water is	Adafruit	Pre calibrated from factory	$10
	Salinity	How salty (conductive) the water is	Vernier	Pre calibrated from factory	$138
Chemical	Total Dissolved Solids (TDS)	How many dissolved solids are in the water	DFRobot	Can be calibrated	$12
	pH	How acidic or basic the water is	Atlast Scientific	Every 3 months	$68

For communication, a 4G module is tested and can provide communication capabilities when the system has sufficient power to enable the connection. A Real Time Clock (RTC) module was used to help keep track of time between sensor readings and know when each reading was taken. This was added in case the system were to ever lose power at any point or needed to restart.

B. Powering the System

A key feature we wanted for this system is for it to be highly flexible with the potential to deploy almost anywhere. In order to achieve this, it cannot rely on receiving power from the grid. Solar panels and a rechargeable battery were chosen as a way to provide continuous low maintenance power. The greater the durability of each component, the less maintenance and exchange of parts that will need to be performed after deployment.

Monocrystalline solar panels are known to have some of the longest life spans within the solar panel market. Although typically bigger and less flexible compared to other options such as Copper Indium Gallium deSelenide (CIGS), the higher efficiencies, durability, tolerance to extreme temperatures and widespread availability of monocrystalline panels are desired features for our system. Specifically, a small and low cost ($36) 25 Watt 12 Volt solar panel was chosen for our initial prototype. The panel does not power the Arduino directly, instead it charges a small battery which allows periodic data collection through the night and on cloudy days.

We chose a small 12 Volt Lithium Iron phosphate (LiFePO4) Battery to power the system due to its ability to function in extreme temperatures and its high resistance to catching fire. This makes it ideal for outdoor applications, especially in hotter and dryer climates where wildfires are most common. These types of batteries however must not be charged in cold environments as this can damage them.

In order to ensure that the battery is being charged properly at the correct voltage, a solar charge converter is needed. We chose a converter specifically made to be compatible with LiFePO4 Batteries.

Finally, we used a step down converter between the battery and the Arduino to prevent damage to the VIN pin from over voltage.

C. Water Proofing

In order to make the system resistant to the elements, all components except for the sensors and solar panel were placed inside a plastic bin with a lid that contained another smaller bin inside of it. Two holes were then drilled on opposite ends of the bigger bin. On one end, the sensor wires were threaded through the hole into the bread board. On the other end, the cable connecting the solar panel to the solar charge converter was threaded through the hole. Both ends were then sealed off with a glob of hot glue to help prevent any dirt or water from entering. Two bins were used so that any water or dirt that still happened to leak in around the edges of the lid would not come into direct contact with any of the Arduino or battery. Figure 2 shows a closeup of our prototype before placing the lid.

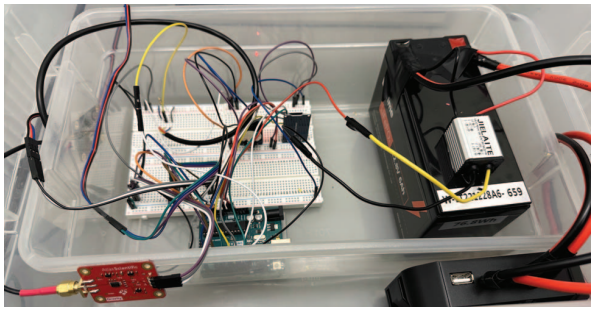


Fig. 2: System Prototype

IV. System Evaluation and Deployment

In this project, we have chosen the Salmon Creek watershed as a testbed. As shown in Figure 3, the Salmon Creek watershed is located in Clark County of southwest Washington, drains an area of approximately 93 square miles immediately north of the city of Vancouver. Salmon Creek originates on the slopes of Elkhorn Mountain (elevation = 2230 ft) and flows approximately 26 river miles to its confluence with Lake River (elevation = 10 ft). The Salmon Creek watershed makes a great testbed for the proposed research, for multiple reasons: 1) The interstate highways and state routes bisect the watershed, which makes it a perfect testbed for studying the short-term and long-term environmental impact of transportation systems; 2) Land use varies throughout the watershed, with commercial timberland and rural residences dominating the upper watershed. Urbanization increases moving downstream through the watershed, with fairly developed commercial and residential areas in the lower watershed. This provides potentials to study the inequitable environmental impacts on different communities; 3) The USGS and Department of Ecology have been studying the water quality of surrounding creeks, rivers, lakes for decades [16]. Historical data are available to public, which makes it possible to develop and validate data analytics algorithms.

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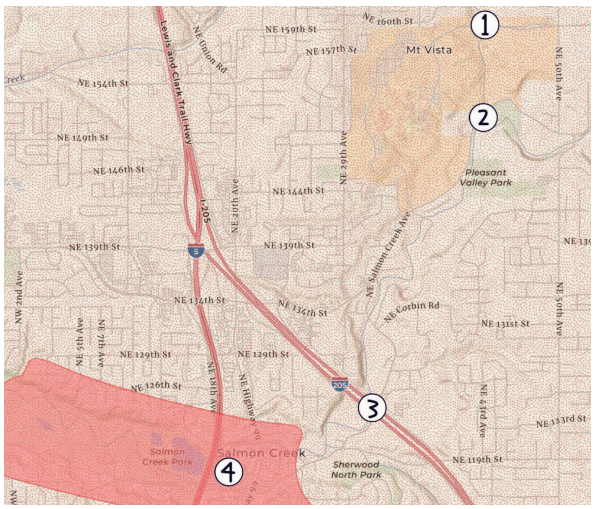


Fig. 3: Data Collection Locations in Salmon Creek Watershed

Figure 3 shows the four different locations within the Salmon Creek watershd which we chose to deploy our system. These locations provide easy access to receiving water and are representative in terms of the distance to major highways. Figure 4 shows our system deployment at one of the locations.

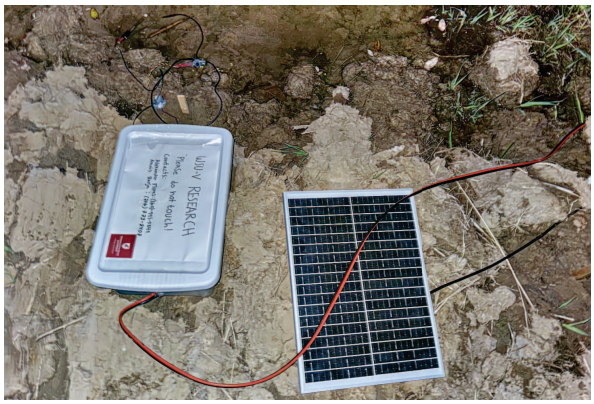


Fig. 4: System Deployment

A. Lab Testing

Before deploying our system, we have validated the system components, especially the multisensing component in a lab setting. In order to find out how vehicles influence water quality, we measured the changes in water after creating a solution of distilled water and select few chemicals commonly used for vehicle maintenance. Additionally, we tested each solution at various temperature (10°C, 22°C, and 50°C) as well as after a week of evaporation in a fume hood. We chose to focus on chemicals used for vehicle maintenance because they are a common form of vehicle pollution that readily get into our water sources.

TABLE II: Common Car Maintenance Chemicals

Car Maintenance Chemicals	
Antifreeze	Fuel Stabilizer
Glass Cleaner	Motor Oil
Multi Purpose Cleaner	Protectant
Starting Fluid	Steering Fluid
Synthetic Brake Fluid	Transmission Fluid
Tire Shine	Washer Fluid Additive
Wax and Wash	Windshield Washer Fluid

Table II shows the 14 chemicals we used in the lab experiments. To test the water quality of all 14 chemicals at various conditions, we performed the following experiment:

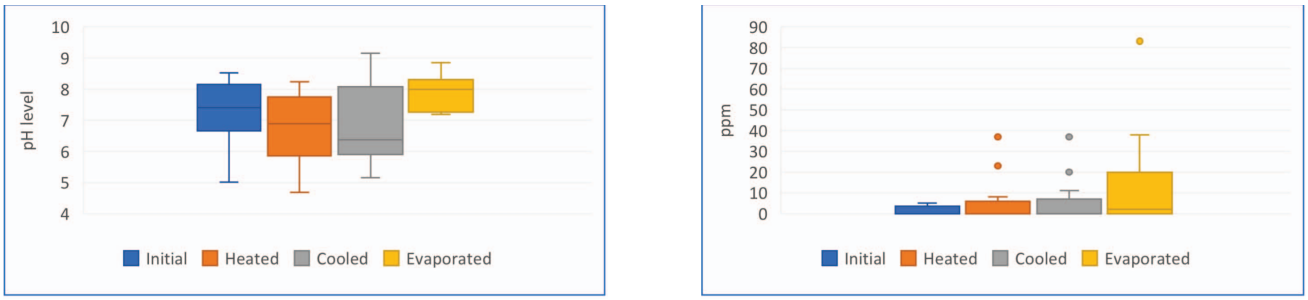
1) Fill a cup with 207 mL of distilled water.
2) Mix the cup of distilled water with 2 mL of one of the chemicals.
3) Place each sensor into the solution and record sensor readings.
4) Remove each sensor and wash each sensor with dish soap.
5) Place the solution inside a cooling chamber and cool it down to 10°C.
6) Remove the solution from the chamber before repeating steps 3 and 4.
7) Place the solution inside a heating chamber and heat it up to 50°C.
8) Remove the solution from the chamber before repeating steps 3 and 4.
9) Place the solution in a fume hood for 1 week.
10) Remove the solution from the fume hood before repeating steps 3 and 4.

TABLE III: Average Car Chemicals Effect on Water Quality in various Conditions

Condition	pH	Temperature (C°)	TDS (ppm)
Initial	7.3	21.6	4.1
Heated	6.7	47.8	5.8
Cooled	6.9	13.6	5.5
Evaporated	7.9	21.7	13.9
Distilled Water	7	25.6	0.0
Average Change	7.2	26.2	7.3

Table III shows the results of the experiments. When comparing the pH and TDS of each solution from the experiment to plain distilled water, the results show that on average each chemical changed the pH by 0.2 and TDS by 7.3 ppm. The pH and TDS distributions are shown in Figure 5a and Figure 5b, illustrating how these chemicals affect the water. Looking at the pH distribution, pH stayed at relatively safe levels (6.5-8.5) for most chemicals and conditions, but a few of the chemicals reached unsafe levels (multi purpose cleaner and fuel stabilizer for most conditions). By analyzing the TDS distribution shown in Figure 5b, we can draw the same conclusion as pH. Specifically, most of the car chemicals contribution zero TDS, while a handful of chemicals contributed small amount of TDS. The top TDS pollutant was wax and wash, followed by antifreeze, for all the conditions. None of the car chemical

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(a) pH

(b) TDS

Fig. 5: Distribution of pH and TDS in Various Conditions

led to unsafe level of TDS (∼350 ppm). Additionally, the TDS distribution showed that most chemicals are not fully soluble in water, which explains the slight changes of pH and TDS, as shown in Table III. Measuring additional water quality parameters would allow us to gain an even bigger picture of how each pollutant effects water quality.

B. Field Testing

To evaluate local water quality, we assessed the water quality of 4 nearby level 1 and 2 critical aquifer recharge areas, as shown in Figure 3. We choose these sites because they are important water resource locations; water seeps into the ground of these sites and refills our local aquifers. Each location is also either very close to a road or connected to a body of water that is. There were also no raised edges or curbs in the area of the road nearest to the water which would allow runoff to more easily enter the water when compared to a road with a curb. Each area did however have road verges which would prevent any run off from flowing in directly into the water without going through some vegetation first. For each location, we collected and analyzed samples once a week for the duration of 1 month.

TABLE IV: Average Water Quality in Salmon Creek after 1 Month

Location	pH	Temperature (C°)	TDS (ppm)
1	8.2	21.5	82.3
2	8.1	21.8	80.0
3	8.1	21.5	73.0
4	8.0	21.9	76.7
Average of all Locations	8.1	21.7	78.0

Table IV shows the measurements collected from the Salmon Creek watershed. The average pH levels are 8.1 and the average TDS are 78 ppm. This suggests that the local surface water contains some pollutants but is still within safe limits.

Determining at a direct pollution source is challenging with limited amount of data. The pH and TDS values indicate that local water is polluted and more data is required for further water quality analysis. Interestingly enough, the locations near the busier higher trafficked roads (Locations 3 and 4) had a lower average of TDS compared with the other 2 locations. One possible explanation for this is that the water with the higher TDS readings flowed less and were overall more static or connected to other static bodies of water. This would allow more pollutants to build up over time in those areas. Multiple water quality readings at different areas of a chosen location could help with determining a more accurate average. Our system provides the potential for long-term automated monitoring, which will eventually help address this challenge.

V. Data Processing and Analysis

Our system is in its early stage of deployment, therefore only limited data is available. However, we have explored the various opportunities with utilizing open-source datasets to gain insights of the water quality issues in the region. Open-source datasets provide us with discrete historical data from our local environment. These datasets originate from two government organizations, the National Oceanic and Atmospheric Administration (NOAA) [17] and the National Water Quality Monitoring Council (NWQMC) [18]. The NOAA dataset provides us with monthly weather data (air temperature, precipitation, snowfall), while the NWQMC dataset provides us with water quality data (pH, conductivity, turbidity, and water temperature).

Both datasets were filtered to remove duplicates, missing values, and irrelevant columns. The data from NWQMC was filtered to only include data similar to our water types, then pivoted so that each water type had its own column. This data was then aggregated with the NOAA data to find relationships between water and weather parameters. They were joined based on the month and year that matched between the monthly weather summaries and when the water quality data from NWQMC was recorded. The data was then split by ground and surface water types.

Figure 6 shows an initial analysis of the NWQMC dataset on the correlation between the water temperature and conductivity. The data collection location is the Columbia River marker #47, or upstream of Willamette River. As shown in the figure, there is a strong correlation between these two parameters. Note that normally these two parameters have a positive relationship but in the case of this dataset their relationship is negative (conductivity is flipped). One potential explanation for the negative correlation is the heavy use

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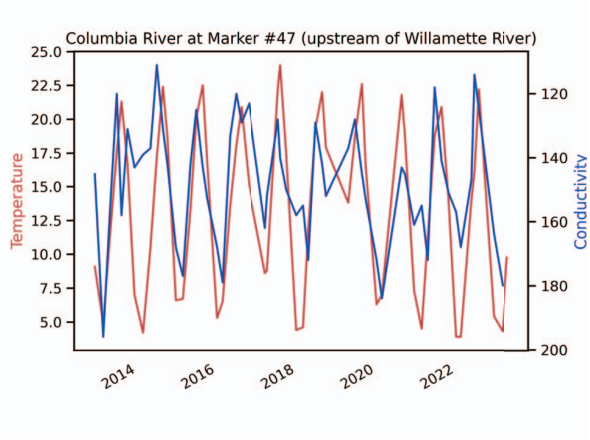


Fig. 6: Correlation between Water Temperature and Conductivity

of deicers (road salts) in winter due to the high influence of salts on conductivity [19]. With continuous monitoring enabled by our system, the root cause of these interesting observations can be validated once sufficient data are collected from the region. Combining our system with the advanced data analytics technologies will open up significant opportunities for understanding the mitigating the negative environmental impacts of the transportation systems.

VI. Conclusion

Transportation systems can affect the surrounding water quality in many ways. Ordinary operation and wear and tear of our vehicles and infrastructure result in a variety of contaminants in our water through wash off from rain or snow storm events. Receiving surface and ground waters are both susceptible to contamination from these sources. Due to the slow movement of ground water through aquifers, it is challenging to fully understand the environmental impact of transportation systems.

In this paper, we present our initial work in enabling long-term automated monitoring of roadside water quality using a portable remote sensing system. The system was evaluated in both the lab setting and the field setting. We have deployed at multiple locations within the Salmon Creek watershed to collect real-world water quality data. Although there have been previous studies that are working on monitoring environmental issues, our system specifically focuses on the potential for long-term understanding of pollution caused by transportation systems. We believe this system will provide a foundation for continuous monitoring and analysis of roadside water quality, leading to more informative decision making for mitigating the negative impacts of transportation systems. For future work, we will perform longer-term data collection, and utilize advanced AI and machine learning algorithms to learn the root causes of various water quality issues.

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